

Ricardo Barrios

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Full-stack developer and Systems Engineering student with expertise in React, Tailwind, Django, Spring Boot, Next.js, and AI/ML solutions. Proven ability to build scalable applications, optimize system performance, and enhance user experience. A strong problem-solver with a growth mindset, highly organized, and driven for precision while remaining adaptable and open to feedback.

Skills

- **Programming Languages:** TypeScript, JavaScript, Java, Python, C++, PHP
- **Libraries & Frameworks:** React, Next.js, Tailwind, Spring Boot, Django, Node.js, TensorFlow, Scikit-learn
- **Tools & Platforms:** Docker, Netlify, Google Cloud, PostgreSQL, MySQL, MongoDB, SQLite, Power BI
- **Languages:** Fluent in Spanish and English, Conversational French (A2-B1)

Selected Projects

Freelance Web Developer, INGINOVA SAS

February 2025 – March 2025 | [Live Demo](#)

- Created INGINOVA's first corporate website to establish a digital presence and enhance its image in the civil engineering sector.
- Utilized technologies like Typescript, Next.js, React, Tailwind, Netlify, and integrated DecapCMS for efficient content management.
- Delivered a performance-optimized solution that enabled non-technical staff to manage content easily, increasing marketing reach and business growth.

Academic Project Lead, AI Mango Classification, Universidad del Magdalena

September 2024 – November 2024 | [Notebook](#)

- Led a team of 4 students to address the challenge faced by agricultural producers in efficiently determining mango ripeness for inventory and quality control.
- Developed multiple classification models, including Deep Neural Networks (DNNs), and applied image preprocessing techniques and data augmentation.
- Created a system that **achieved 95%+ accuracy**, demonstrating the practical application of AI and machine learning in optimizing the agricultural supply chain.

Personal Project Developer, Habit Tracking System

August 2024 – October 2024 | [Live Demo](#)

- Identified the need for an accessible habit tracking solution that functions without requiring internet connectivity.
- Designed and built a React-based PWA with an offline-first architecture and implemented a gamified reward system to boost motivation.
- Created a fully functional application with persistent local storage, enhancing personal productivity and enabling users to share the app for broader use.

Academic Project Lead, Electronic Voting System, Universidad del Magdalena

March 2024 – May 2024 | [Repository](#)

- Addressed issues with traditional paper ballots in school elections, such as security, voter privacy, and inefficiency in the process.
- Engineered frontend and backend components for user authentication, email notifications, and database interactions, and implemented role-based access control to secure election administration.
- Enabled a secure, efficient election process with restricted access, improving voter privacy and ensuring only authorized users could access sensitive functions.

Freelance Web Developer, Legal Education Platform

April 2023 – June 2023

- A group of law students approached me to create a platform to connect with tutors for personalized learning.
- Built a Django-based web application with real-time booking capabilities and session management.
- Delivered a platform that improved access to legal education through personalized tutoring, with secure authentication and scheduling features.

Education

Bachelor of Science in Systems Engineering

Universidad del Magdalena, Santa Marta, Colombia | Expected July 2026 | CGPA 4.2/5.0

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Software Architecture, Ethical Hacking, Web Development, Microservices, User Interface Design, Networks, Data Analytics with Power BI.